

DEGREE: PHYSICS

EXPERIMENTAL PROYECT II

EXPERIMENT 5

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1 Introduction

In this experiment we are going to find equipotential field lines between 2 electrodes in 3 different experiments with different electrodes and know if they are lines or surfaces, we're going to know as well how works the Electric field in the equipotential lines an how is it calculated.

2 Principles

The equipotential lines and surfaces are used to represent the shape of the electric fields generated by electrical charges. If a charge moves along these equipotential lines or surfaces, no work is done because the potential remains the same at every point on each line or surface. Knowing that the lines or surfaces of equal potential are perpendicular to the electric field lines, we can determine the electric field lines by experimentally identifying the positions of the equipotential lines. The spatial arrangement or shape of the electric field generated by charges determines the form that the equipotential lines take. When we talk about an isolated point charge, the formula for electric potential (V) is directly dependent on the charge (Q) and inversely dependent on the radial distance from the charge (r):

$$W = \frac{kQ}{r} \quad (1)$$

We can say that there is a radial dependence because the potential remains the same at any point that is equidistant from the point charge. Therefore, the equipotential lines for a single point charge are circular, with the point charge at the center.

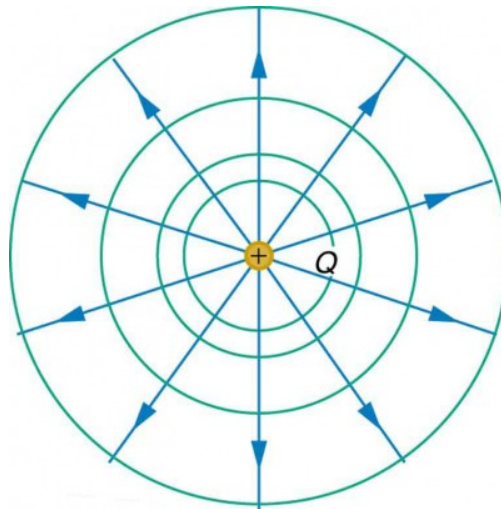


Figure 1: Single point charge.

However, when multiple discrete charges interact, their electric fields overlap. This results in a combination of fields, and in areas close to both charges, the effects of both fields are observed. On the other hand, in the case of charges distributed across two conducting plates in static equilibrium, there will not be any significant changes in the equipotential lines.

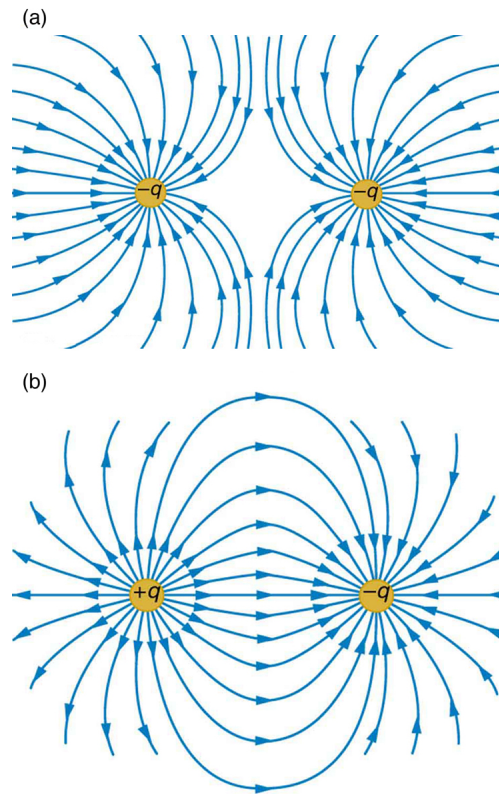


Figure 2: Interaction.

Finally, this results in a continuity of effects at any point located at a certain distance from either plate.

3 Experimental Procedure

To carry out the experiment, first of all we must set it up. The assembly consists of a transparent box that we will fill with distilled water, a current that will be connected through two pairs of safety experiment leads, to some electrodes (rectangular and circular) as well as a multimeter.

When we have all the material prepared, we connect the current to one electrode and the other electrode to a voltmeter, which will calculate the potential difference between the two electrodes. We fill the transparent box with 400cm^3 of distilled water and place a squared sheet with the electrodes drawn under the box. Next we place the two electrodes inside the water in the space that is drawn for them on the sheet. Finally we will only have to pass the multimeter through the marked crosses and write down the value that the voltmeter shows us.

We have to follow the same steps for all types of electrodes, first we will use the two rectangular ones and measure the entire measurement sheet, then we will do the same with the circular ones and finally we will place a hollow circular barrier in the center of the sheet with the two circular electrodes at the tips and we will measure the sheet again.

Here we have a photograph of the setup that we will use:

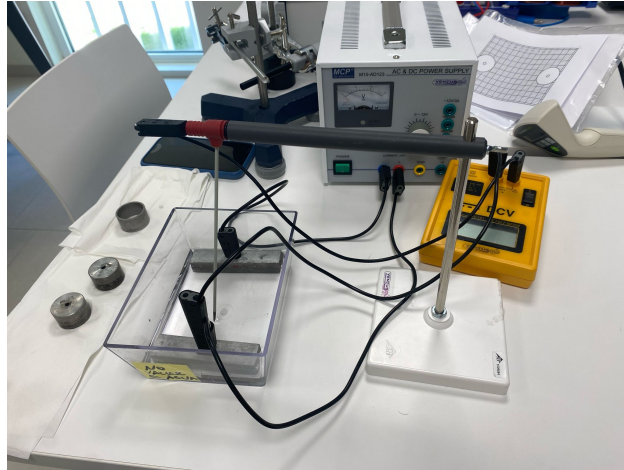


Figure 3: Experimental setup.

4 Calculations

We want to describe the behavior of the electronic potential in the 3 cases mentioned before. For the rectangular one, we obtain this graphic:

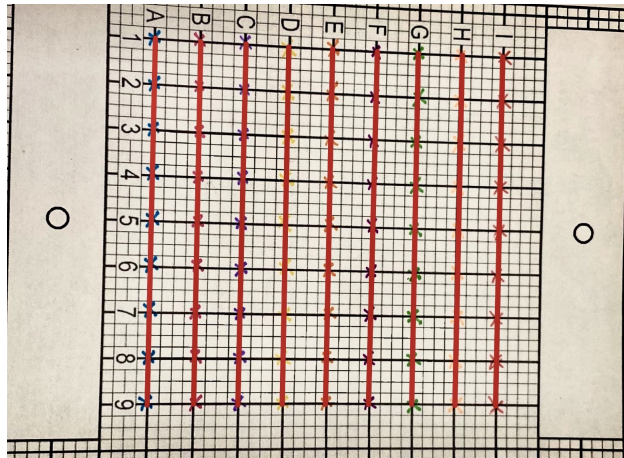


Figure 4: Equipotential lines with rectangular electrodes.

We can see that the equipotential lines have the same value in a row. So the function that describes this is constant.

$$y_A = 3[V], \quad y_B = 2.7[V], \quad y_C = 2.5[V], \quad y_D = 2.15[V], \quad y_E = 1.85[V]$$

$$y_F = 1.55[V], \quad y_G = 1.25[V], \quad y_H = 0.95[V], \quad y_I = 0.65[V]$$

As we can see, the values of y are decreasing in 0.3 ± 0.05 due to a systematic error of the machine.

Now we have the circular electrodes, we have this graphic:

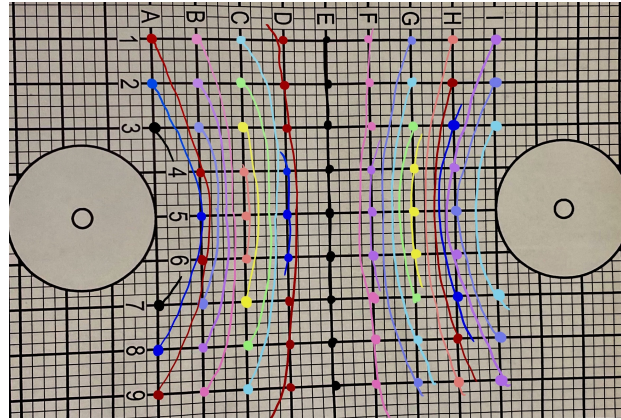


Figure 5: Equipotential lines with circular electrodes.

The lines that are created by the same Voltage points are given by the circle equation. As the values are not that easy as before, we are going to show a table with all the values measured.

	A	B	C	D	E	F	G	H	I
1	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75
2	0.7	0.95	1.2	1.5	1.75	2	2.3	2.6	2.8
3	0.55	0.9	1.15	1.5	1.75	2	2.35	2.7	3
4	-	0.75	1.1	1.45	1.75	2.05	2.4	2.75	-
5	-	0.7	1.1	1.45	1.75	2.05	2.4	2.8	-
6	-	0.75	1.1	1.45	1.75	2.05	2.4	2.75	-
7	0.55	0.9	1.15	1.5	1.75	2	2.35	2.7	3
8	0.7	0.95	1.2	1.5	1.75	2	2.3	2.6	2.8
9	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75

Figure 6: Data with circular electrodes.

Finally we have the one with the circular barrier, it's quite similar to the circular one since it has the same electrodes so the equipotential lines are the same, but in the points inside of the barrier we can observe that they have the same potential value with an error, as we said before, of ± 0.05 . We have the same table as before changing some values:

	A	B	C	D	E	F	G	H	I
1	0,70	0,95	1,20	1,45	1,70	1,95	2,20	2,45	2,70
2	0,65	0,90	1,15	1,45	1,70	1,95	2,25	2,55	2,75
3	0,50	0,85	1,10	1,45	1,70	1,95	2,30	2,65	2,95
4	-	0,70	1,05	1,80	1,85	1,85	2,35	2,70	-
5	-	0,65	1,05	1,80	1,85	1,85	2,35	2,75	-
6	-	0,70	1,05	1,80	1,80	1,80	2,35	2,70	-
7	0,50	0,85	1,10	1,45	1,70	1,95	2,30	2,65	2,95
8	0,65	0,90	1,15	1,45	1,70	1,95	2,25	2,55	2,75
9	0,70	0,95	1,20	1,45	1,70	1,95	2,20	2,45	2,70

Figure 7: Data with circular electrodes and barrier.

As you imagine, the data more bright are inside of the barrier, we can either see that the rest of the values have decreased by ± 0.05 , here we have an image of the barrier with the circular barrier:

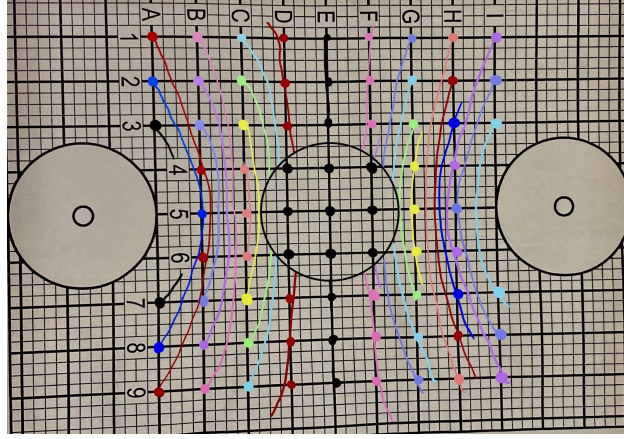


Figure 8: Equipotential lines with circular electrodes and barrier.

Now we have to calculate the value of the electric field. This one is calculated by:

$$E = \frac{\Delta V}{\Delta d}$$

Depending on whether we are in the rectangular or circular electrodes example, we talk always about the distance, the difference between one and the other equipotential line; in the case of the circular, we talk about the radius and the rectangular just distance.

For the error, we can derive partially the electric field between the 2 variables and we have:

$$\frac{\partial E}{\partial V} = \frac{1}{\Delta d}, \quad \frac{\partial E}{\partial d} = -\frac{\Delta V}{\Delta d^2}$$

So we have:

$$dE = \frac{\partial E}{\partial V} dV + \frac{\partial E}{\partial d} dd$$

We substitute the last derivatives and we have the following expressions:

$$\Delta E \approx \left| \frac{1}{\Delta d} \right| \Delta V + \left| -\frac{\Delta V}{\Delta d^2} \right| \Delta d$$

$$\Delta E = \frac{\Delta V}{\Delta d} + \frac{\Delta V}{\Delta d^2} \Delta d$$

Factoring and approximating for minimum errors:

$$\Delta E = E \sqrt{\left(\frac{\Delta V_{Error}}{\Delta V_{Difference}} \right)^2 + \left(\frac{\Delta d_{Error}}{\Delta d_{Difference}} \right)^2}$$

Now we know that our error related to the Voltage is: $V = \pm 0.05[V]$ and related to the distance/radius: $d = \pm 0.001[m]$. We already know that the difference between 2 equipotential lines is approximated $\Delta V = 0.3$ and the distance between one and the other line is $\Delta d = 1[cm] = 0.01[m]$. So we calculate E:

$$E = \frac{\Delta V}{\Delta d} = \frac{0.3[V]}{0.01[m]} = 30V/m$$

So finally we calculate the error:

$$\Delta E = 30 V/m \cdot \sqrt{\left(\frac{0.05}{0.3}\right)^2 + \left(\frac{0.001}{0.01}\right)^2}$$

$$\Delta E = 30 V/m \cdot \sqrt{(0.1667)^2 + (0.1)^2}$$

$$\Delta E = 30 V/m \cdot 0.194$$

$$\Delta E \approx 5.8 V/m$$

So now we can express the electric field like:

$$E = 30 \pm 5.8[V/m]$$

5 Discuss and evaluation

1. **Is it possible to determine the shape of the electric field for each electrode configuration based on the collected data?**

Yes, we know that the electric field is always perpendicular to the equipotential surface, so we can use this equation to make an idea of how can be the shape of these field. Using the values that we have observed, we use this equation

$$E = -\nabla V$$

knowing E as the electric field and V as the voltage.

2. **Can the experimental method used determine equipotential lines in empty space?**

We can use the same method but the results can't be the same because of two reasons. The effect of the water electrical permittivity is not the same as the empty electrical permittivity. In an empty space there is no effect of the polarization of the water, being one of the aspects more modified of the electric field.

3. **In the configuration of parallel plates, in what direction with respect to the equipotential lines was the greatest potential difference measured, and in which directions does the electric field go?**

The greatest potential differences measured was in the row A with a voltage of $3[V]$. The direction that follows is The electric field direction is from the positive plate to the negative plate.

4. **What type of shape do the equipotential lines have when the point-charge-bar configuration was used?**

The equipotential lines have lineal shape that is due to the big amount of charges in the electrode that is quite infinite so the equipotential lines will be horizontal and parallel each other in the end it will be like an ellipse but what we observe is parallel lines.

5. **Were equipotential lines or surfaces detected in this experiment?**

In this experiment we're detecting equipotential surfaces because if we zoom out the two cases (the rectangular and the circular one), we can see that the lines surrounds the electrodes taking a similar shape as them. In both cases we can represent in 2D (our case) the same result as in 3D (equipotential surfaces) like equipotential lines.

6. **How does the mineral concentration affect the experiment?**

The mineral concentration can affect in several ways to the measurement of the equipotential lines. On one hand the mineral concentration includes salts and ions, these ones increases the conductivity and the concentration (stronger or lower electric field), in conclusion, the mineral concentration distort the measurement creating wrong Voltage values for that reason we use distilled water.

6 Bibliography

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Wikipedia. (2021, May 1). *Superficie equipotencial*. Wikipedia. https://es.wikipedia.org/wiki/Superficie_equipotencial